

# Robustness Through Collective Intelligence



**Invited Speaker**

**Ren Wang**

*Illinois Institute of Technology*

**Date:** Mar 3, 2026 (Tue)

**Time:** 11:00am (HKT)

**Zoom Meeting:** 801 137 0362

## Biography

Ren Wang is an Assistant Professor in the Department of Electrical and Computer Engineering at the Illinois Institute of Technology and a faculty member of the Institute for Data, Econometrics, Algorithms, and Learning. His research focuses on Trustworthy Machine Learning, Population-Based Machine Learning, and applications of ML in smart grids, biology, and healthcare. His long-term vision is to develop next-generation trustworthy and intelligent machine learning systems that accelerate progress in engineering and scientific discovery. Dr. Wang has published widely in top-tier conferences and journals spanning machine learning, signal processing, computer vision, control, power systems, and bioinformatics, and has served as an area chair for several premier conferences. He received the 2023 ORAU Ralph E. Powe Junior Faculty Enhancement Award and, as Principal Investigator, has led multiple research projects supported by U.S. federal agencies such as the NSF and DoE.

## Abstract

Adversarial training typically views robustness as a zero-sum game, where securing a model against one attack vector often compromises generalization to others or degrades clean accuracy. This talk proposes a shift from monolithic defense to collective intelligence, introducing a population-based framework for building resilient deep models. I will first present Efficient Robust Mode Connectivity (ERMC), which demonstrates that models robust to disparate  $L_p$  attacks can be unified via continuous, low-loss paths in the weight space, allowing for a shared, multi-norm defense. I will then discuss the Dual-Model Mixture-of-Experts, an architecture that combines clean and robust experts within an architecture to balance accuracy and robustness. Together, these works illustrate that population diversity, spanning both attack types and architectural specializations, enables a level of systemic resilience that exceeds the capabilities of any single, isolated model.